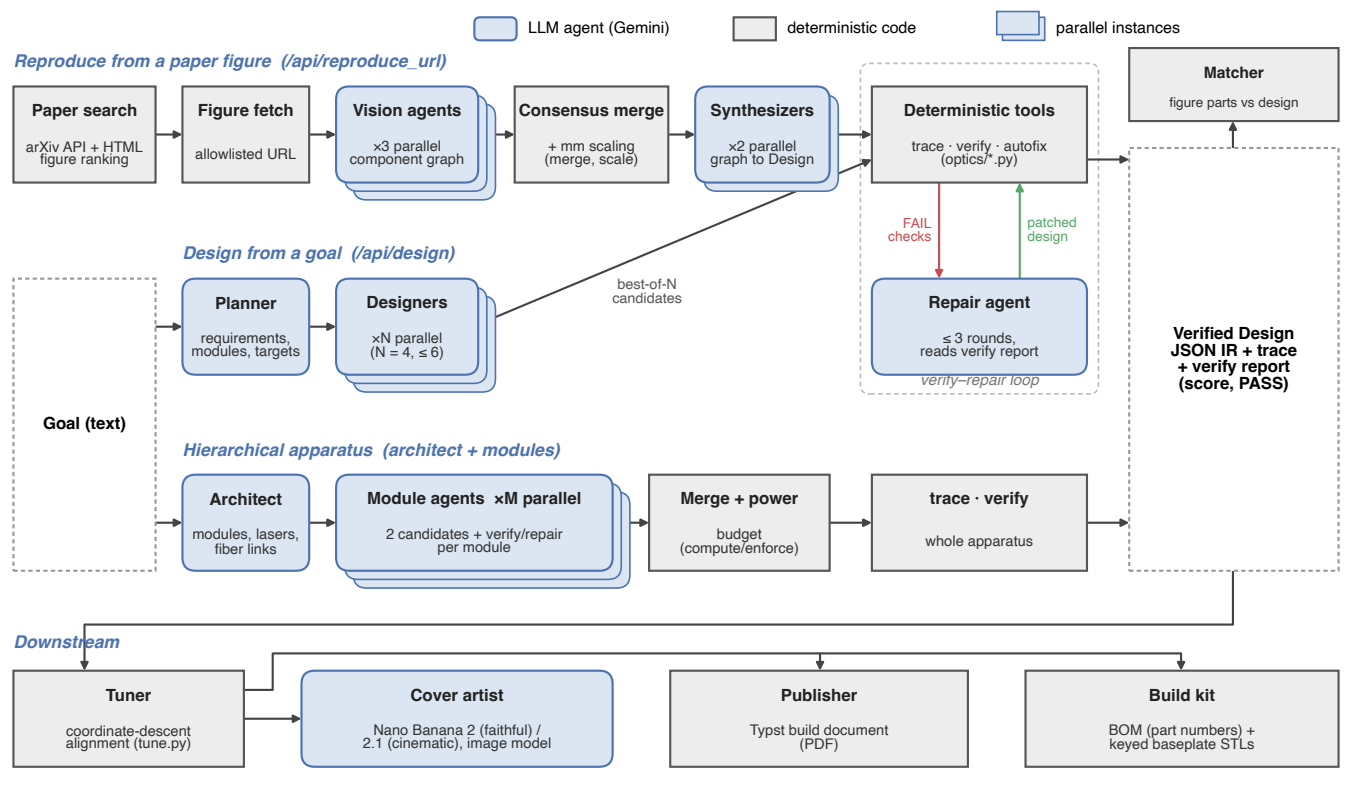
LLM agent (Gemini)
deterministic code
parallel instances
Reproduce from a paper figure (/api/reproduce_url)
Paper search
arXiv API + HTML figure ranking
Figure fetch
allowlisted URL
Vision agents
×3 parallel component graph
Consensus merge
+ mm scaling (merge, scale)
Synthesizers
×2 parallel graph to Design
Deterministic tools
trace · verify · autofix (optics/*.py)
Matcher
figure parts vs design
FAIL checks
patched design
Repair agent
≤ 3 rounds, reads verify report
verify–repair loop
Design from a goal (/api/design)
best-of-N candidates
Goal (text)
Planner
requirements, modules, targets
Designers
×N parallel (N = 4, ≤ 6)
Verified Design
JSON IR + trace
+ verify report
(score, PASS)
Hierarchical apparatus (architect + modules)
Architect
modules, lasers, fiber links
Module agents ×M parallel
2 candidates + verify/repair per module
Merge + power
budget (compute/enforce)
trace · verify
whole apparatus
Downstream
Tuner
coordinate-descent alignment (tune.py)
Cover artist
Nano Banana 2 (faithful) / 2.1 (cinematic), image model
Publisher
Typst build document (PDF)
Build kit
BOM (part numbers) + keyed baseplate STLs